



life.augmented

STM32H5 Security workshop Tools installation

Tomas Team

Tools requirements

- HW Requirements :
 - PC with Windows 10 or upper
 - USB Type C cable
 - STM32H573I-DK will be delivered by ST during the Workshop.
- SW Requirements :
 - [STM32CubeMX](#) version 6.9.2 or upper
 - [STM32CubeH5](#) version 1.1.1
 - [STM32TRUSTEE-SM](#) version 1.1.1
 - [STM32CubeIDE](#) version 1.13.2 or upper
 - [STM32CubeProgrammer](#) version 2.14.0 or upper
 - [Material for hands-on](#)
 - Terminal application (TeraTerm)

Tools installation : STM32CubeMX

- [STM32CubeMX](#) version 6.9.2 or upper
- Download and install the package.

Tools installation

- SW Requirements :
 - [STM32CubeMX](#) version 6.9.2 or upper 
 - [STM32CubeH5](#) version 1.1.1
 - [STM32TRUSTEE-SM](#) version 1.1.1
 - [STM32CubeIDE](#) version 1.13.2 or upper
 - [STM32CubeProgrammer](#) version 2.14.0 or upper
 - [Material for hands-on](#)
 - Terminal application (TeraTerm)



Tools installation : STM32H5Cube

Launch STM32CubeMX and Login

MX STM32CubeMX Untitled

File Window Help

1 myST Login Alt-L

Home

Existing Projects

Recent Opened Projects

Test_bank2.ioc	MX
Last modified date : 31/08/2023 8:42:31	
Test_bank1.ioc	MX
Last modified date : 31/08/2023 8:31:26	
test.ioc	MX
Last modified date : 30/08/2023 10:34:00	
Tamper_detection.ioc	MX
Last modified date : 29/08/2023 15:01:03	
Last_test.ioc	MX
Last modified date : 03/08/2023 15:13:46	

Other Projects

[Data collection information notice](#)

New Project

I need to :

- Start My project from MCU
ACCESS TO MCU SELECTOR
- Start My project from ST Board
ACCESS TO BOARD SELECTOR
- Start My project from Example
ACCESS TO EXAMPLE SELECTOR

Manage software installations

Check for STM32CubeMX and embedded softwa...

CHECK FOR UPDATES

Install or remove embedded software packages

INSTALL / REMOVE

FreeRTOS is back for latest STM32 series now available as a STM32Cube expansion

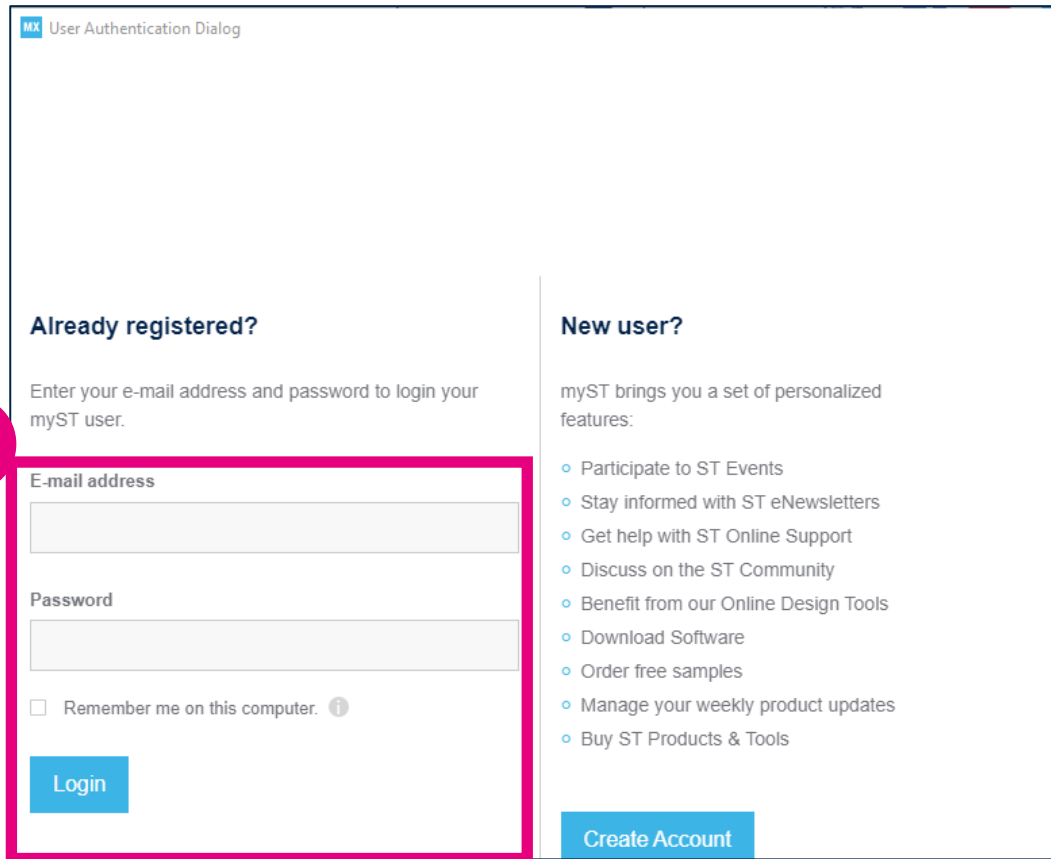
Pinout & Configuration

About STM32 External Tools



Tools installation : STM32H5Cube

Enter your ST account credential or create an account



MX User Authentication Dialog

Already registered?

Enter your e-mail address and password to login your myST user.

E-mail address

Password

☐ Remember me on this computer. ⓘ

Login

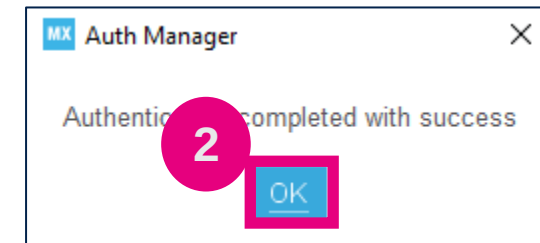
New user?

myST brings you a set of personalized features:

- Participate to ST Events
- Stay informed with ST eNewsletters
- Get help with ST Online Support
- Discuss on the ST Community
- Benefit from our Online Design Tools
- Download Software
- Order free samples
- Manage your weekly product updates
- Buy ST Products & Tools

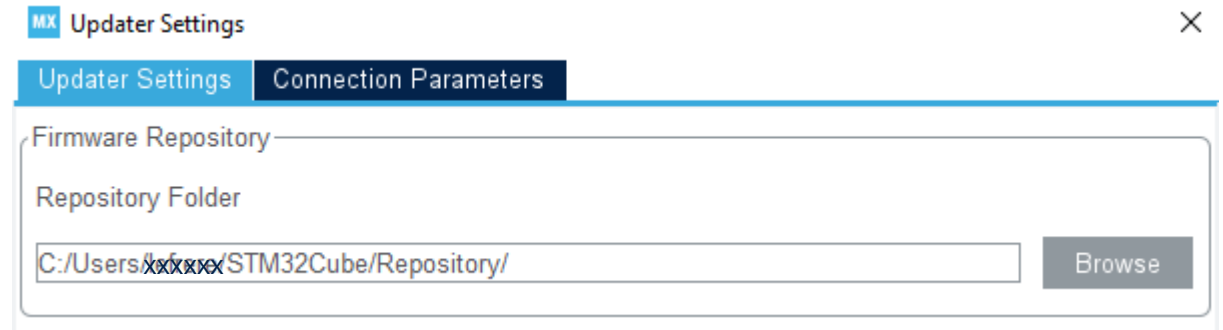
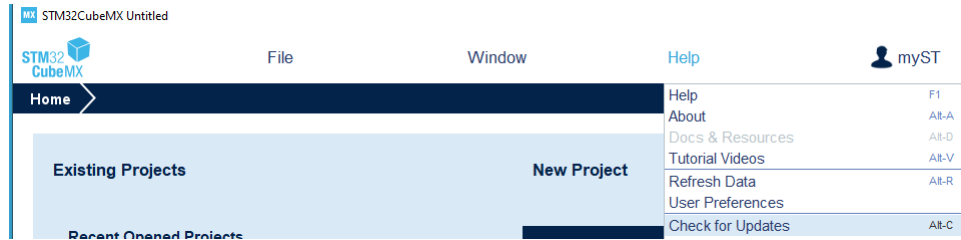
Create Account

A pink box highlights the login fields (E-mail address, Password, Remember me, and Login button) with a pink circle containing the number 1 next to it.





Issue with space in repository path...



- By default the repository is :
C:\Users\<<user name>\lefrere\STM32Cube\Repository\
- If you have space in you <user name>, you will have issue with Secure Manager.
- Please modify the Repository folder to avoid space. If not possible, go on installation and as workaround once installed (slide 12) copy STM32Cube_FW_H5_V1.1.1 to a location without space
For example :
C:\Training\STM32H5_Workshop



Tools installation : STM32H5Cube

Select Help->Manage embedded software package

The screenshot shows the STM32CubeMX application interface. The top menu bar includes File, Window, and Help. The Help menu is open, and the 'Refresh Data' option is highlighted with a red box and a red circle containing the number 1. Below the menu, the 'Existing Projects' section lists recent projects. The 'Manage software installations' section on the right includes a 'CHECK FOR UPDATES' button and an 'INSTALL / REMOVE' button. A central panel titled 'I need to:' offers options to start a project from MCU, ST Board, or Example, each with a corresponding 'ACCESS TO' button.

STM32CubeMX

File Window Help Hello Frantz

Home

Existing Projects

Recent Opened Projects

Test_bank2.ioc MX
Last modified date : 31/08/2023 8:42:31

Test_bank1.ioc MX
Last modified date : 31/08/2023 8:31:26

test.ioc MX
Last modified date : 30/08/2023 10:34:00

Tamper_detection.ioc MX
Last modified date : 29/08/2023 15:01:03

Last_test.ioc MX
Last modified date : 03/08/2023 15:13:46

Other Projects

Data collection information notice

Help F1
About Alt-A
Docs & Resources Alt-D
Tutorial Videos Alt-V
Refresh Data Alt-R
User Preferences
Check for Updates Alt-C
Manage embedded software packages Alt-U
Updater Settings ... Alt-S

1

I need to :

Start My project from MCU
ACCESS TO MCU SELECTOR

Start My project from ST Board
ACCESS TO BOARD SELECTOR

Start My project from Example
ACCESS TO EXAMPLE SELECTOR

Manage software installations

Check for STM32CubeMX and embedded software packages

CHECK FOR UPDATES

Install or remove embedded software packages

INSTALL / REMOVE

MX updated in v8.0
Starting with STM32H5
Define and configure your boot strategy

Project Manager > BootPath Settings

About STM32 External Tools



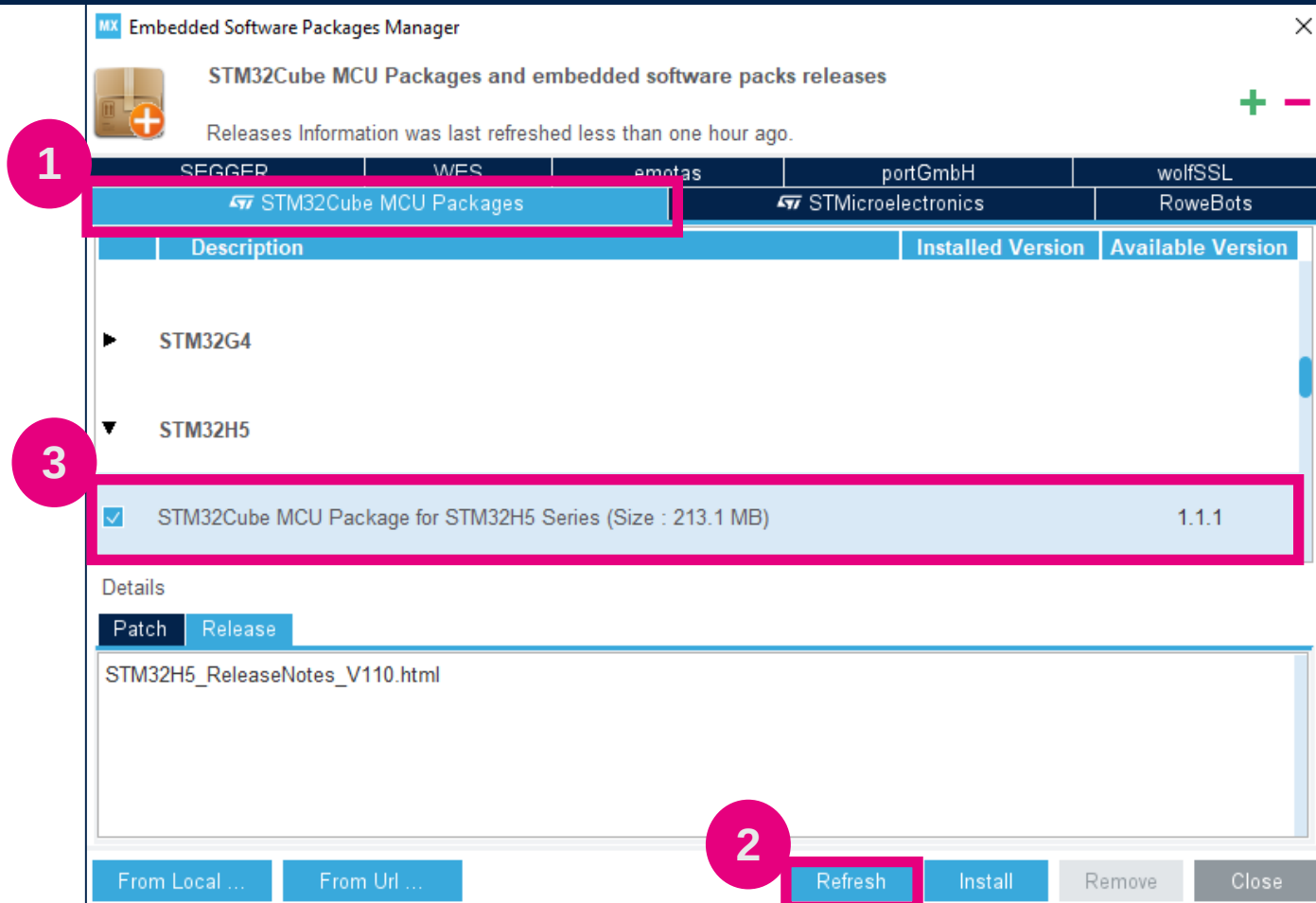
Tools installation : STM32H5Cube

Select Help->Manage embedded software package

The screenshot displays the STM32CubeMX application window. The top menu bar includes 'File', 'Window', and 'Help'. The 'Help' menu is open, showing options like 'Help', 'About', 'Docs & Resources', 'Tutorial Videos', 'Refresh Data', 'User Preferences', 'Check for Updates', 'Manage embedded software packages', and 'Updater Settings...'. A red circle with the number '1' highlights the 'Manage embedded software packages' option. The main interface is divided into two panes. The left pane, titled 'Existing Projects', lists 'Recent Opened Projects' (Test_bank2.ioc, Test_bank1.ioc, test.ioc, Tamper_detection.ioc, Last_test.ioc) and 'Other Projects'. The right pane, titled 'Manage software installations', contains a 'CHECK FOR UPDATES' button, an 'INSTALL / REMOVE' button, and a section for 'Starting with STM32H5' with a 'Project Manager > BootPath Settings' link. The bottom of the window features a footer with 'About STM32' and 'External Tools' links.

Tools installation : STM32H5Cube

Select Help->Manage embedded software package



1

3

2

MX Embedded Software Packages Manager

STM32Cube MCU Packages and embedded software packs releases

Releases Information was last refreshed less than one hour ago.

Description	Installed Version	Available Version
STM32G4		
STM32H5		
STM32Cube MCU Package for STM32H5 Series (Size : 213.1 MB)		1.1.1

Details

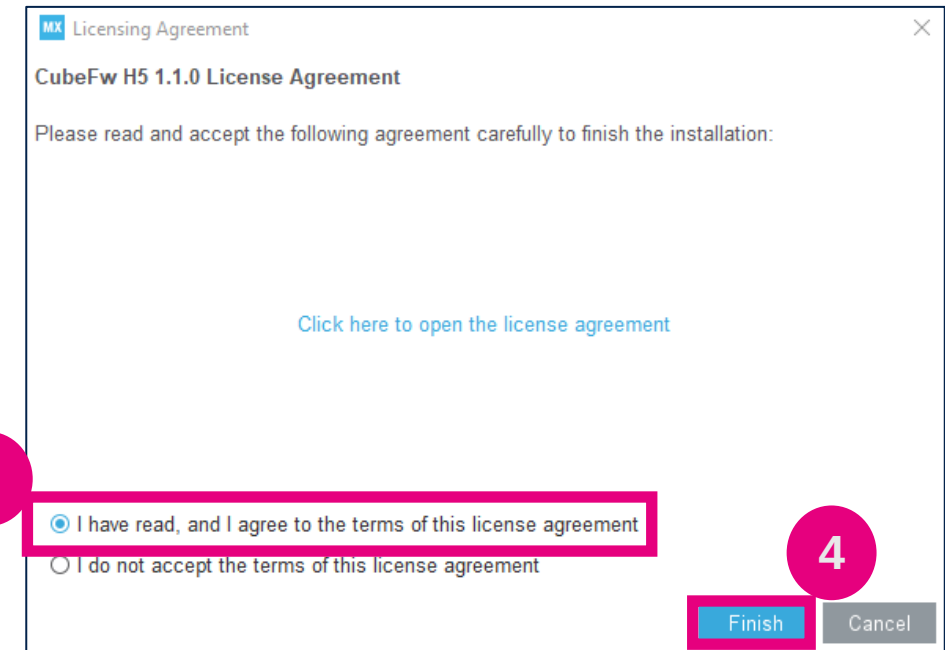
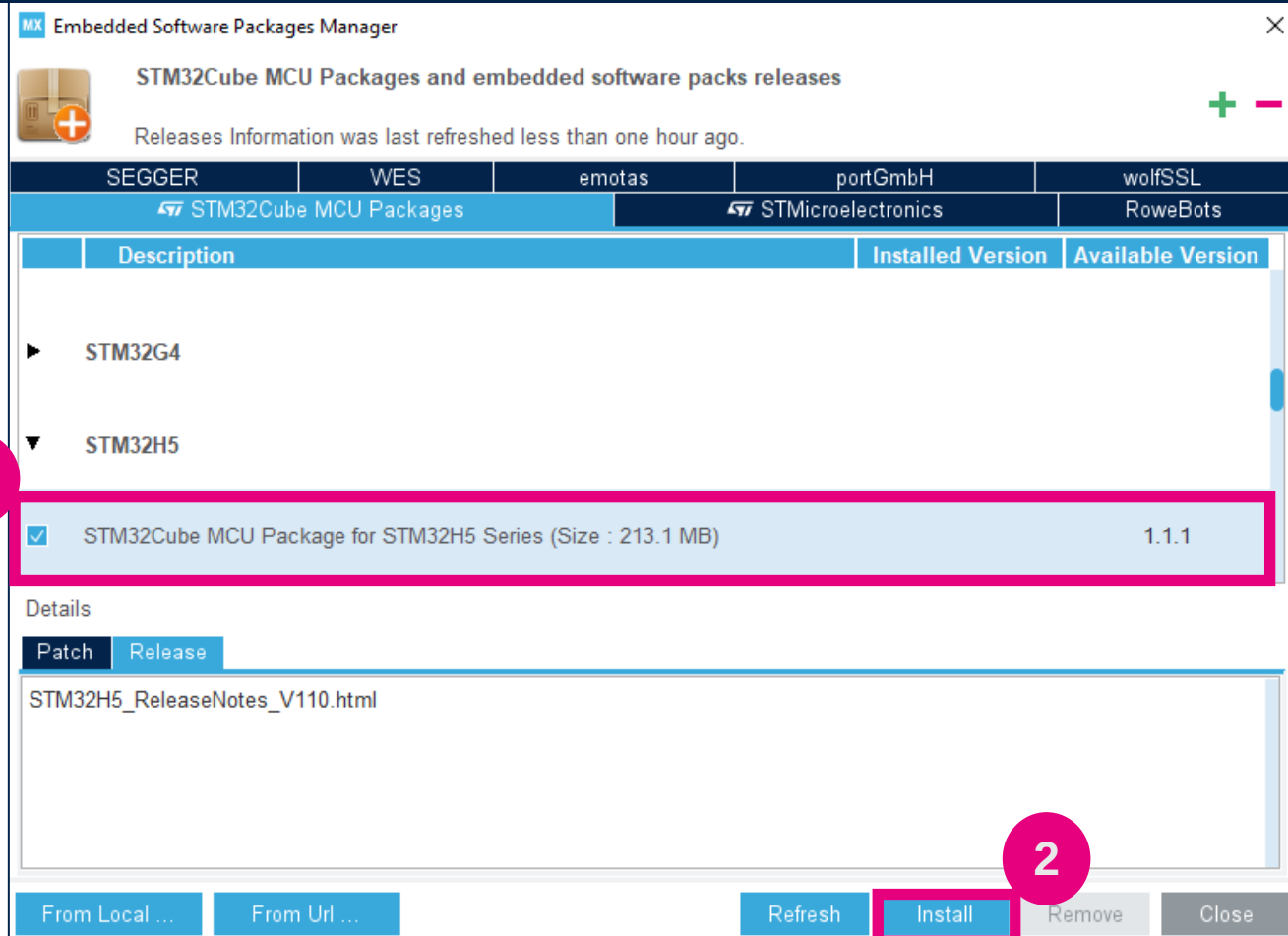
Patch Release

STM32H5_ReleaseNotes_V110.html

From Local ... From Url ... Refresh Install Remove Close

Tools installation : STM32H5Cube

Select Help->Manage embedded software package





Tools installation : STM32H5Cube

Package is installed

MX Embedded Software Packages Manager

STM32Cube MCU Packages and embedded software packs releases

Releases Information was last refreshed less than one hour ago.

SEGGER	WES	emotas	portGmbH	wolfSSL
STM32Cube MCU Packages	STMicroelectronics	RoweBots		

Description	Installed Version	Available Version
▶ STM32G4		
▼ STM32H5		
■ STM32Cube MCU Package for STM32H5 Series	1.1.1	1.1.1

Details

From Local ... From Url ... Refresh Install Remove Close

Tools installation : Secure Manager

Request the secure manager

- Available here <https://www.st.com/en/embedded-software/stm32trustee-sm.html>

Get Software

Part Number	General Description	ECCN (US)	Download
+ X-CUBE-SEC-M-H5	Secure manager embedded software for STM32Cube	5D002.c.1	Request Software

- Request software and fill the form

Tools installation

- SW Requirements:
 - [STM32CubeMX](#) version 6.9.2 or upper 
 - [STM32CubeH5](#) version 1.1.1 
 - [STM32TRUSTEE-SM](#) version 1.1.1
 - [STM32CubeIDE](#) version 1.13.2 or upper
 - [STM32CubeProgrammer](#) version 2.14.0 or upper
 - [Material for hands-on](#)
 - Terminal application (TeraTerm)

Tools installation : Secure Manager

Fill the form

Request Software

Project Title:	
Project Description:	
Application:	Please choose an application
End Application:	Please choose a sub-application
Nature of Business:	
Military Related:	
Country/Region where the Software will be used:	
Please keep me informed about future updates for this product.	☐
Comment:	

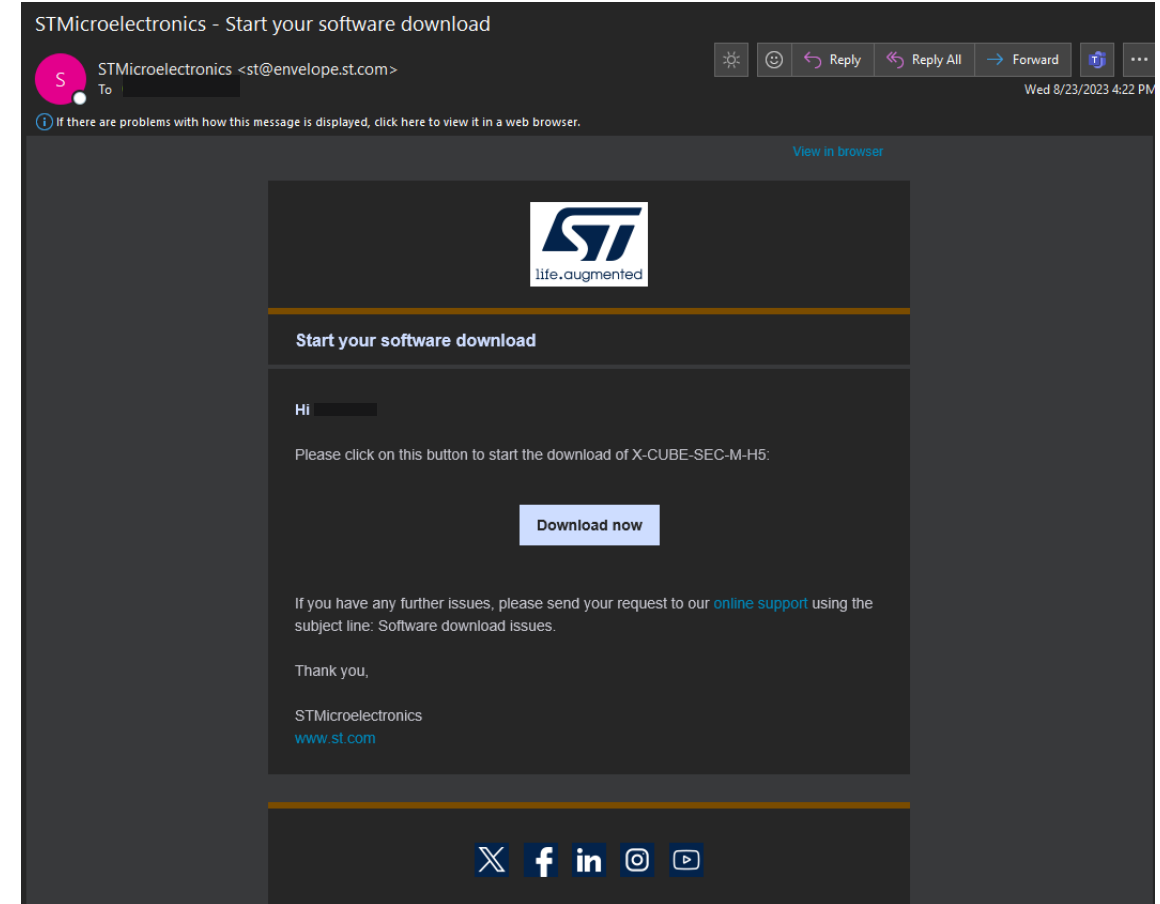
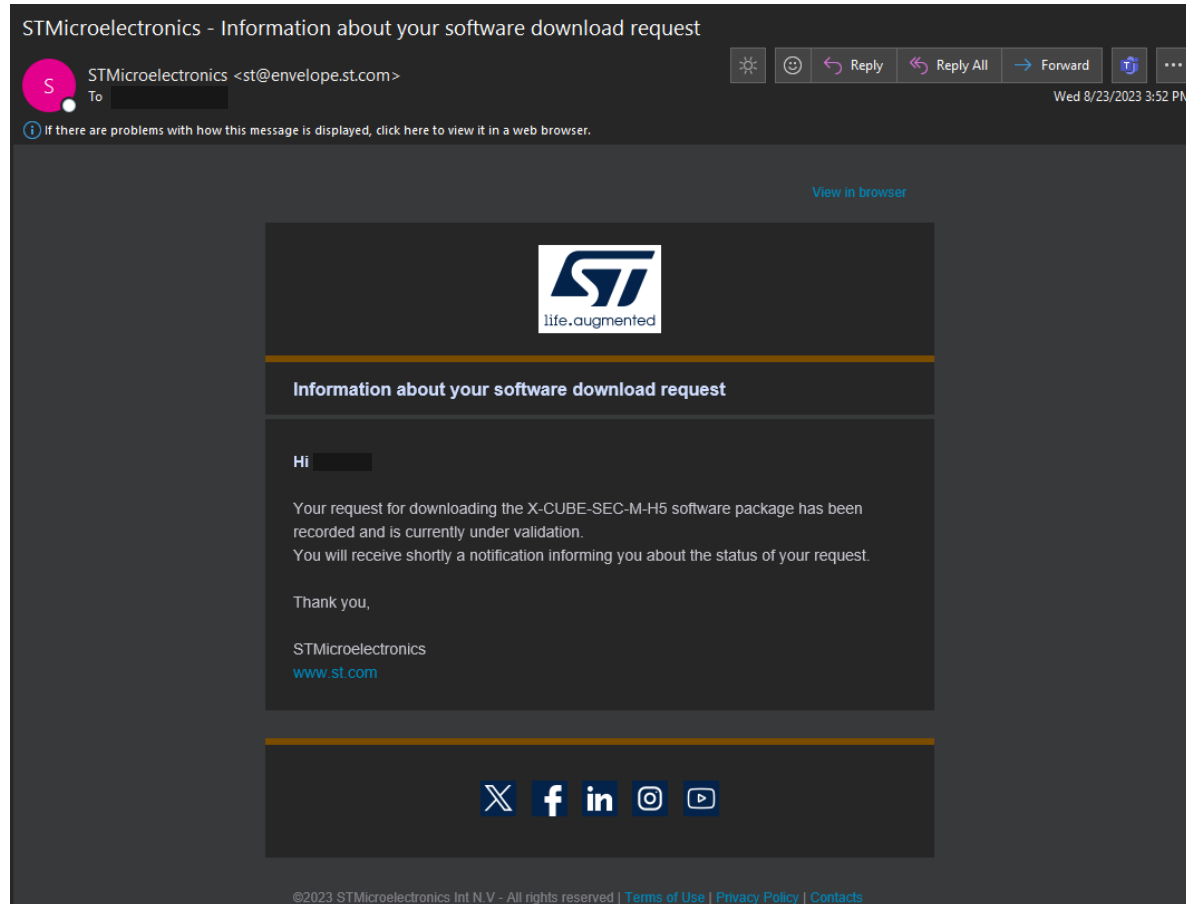
☐ I accept all [Terms & Conditions](#) of the Export Control regulations

Cancel

Confirm Request

Tools installation : Secure Manager

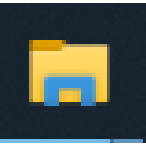
Check your mail box ! You will receive a download link within 24h



Tools installation : Secure Manager

Content of en.x-cube-sec-m-h5-v1-1-1.zip

```
X-Cube-SEC-M-H5_V1.1.1
├── Additional_License_Terms_SMAK.html
├── Package_license_SMAK.html
├── Release_Notes_SMAK.html
├── Projects
│   ├── STM32H573I-DK
│   │   ├── ROT_Provisioning
│   │   │   ├── SM
│   │   │   │   └── Binary
│   │   │   │       ├── SecureManagerCore_PROD_FWU_v1.1.0.bin
│   │   │   │       ├── SecureManagerPackage_PROD_v1.1.1.ssfi
│   │   │   │       ├── STuRoT_FWU_v1.1.1.bin
│   │   │   │       └── version.bat
│   └── _htmresc
│       ├── Add_button.svg
│       ├── favicon.png
│       ├── mini-st_2020.css
│       ├── Security_Secure_Manager.png
│       ├── st_logo_2020.png
│       └── Update.svg
```



Tools installation : Secure Manager

Unzip en.x-cube-sec-m-h5-v1-1-1.zip on STM32Cube_FW_H5_V1.1.1

The image shows a file explorer window with the address bar set to `C:\Users\ xxxx \STM32Cube\Repository\STM32Cube_FW_H5_V1.1.1\`. The file list includes:

- _htmresc
- Documentations
- Drivers
- Middlewareares
- Projects
- Utilities
- package.xml
- Package_license.html
- Package_license.md
- Release_Notes.html

A pink arrow labeled **COPY** points from the source folder `X-Cube-SEC-M-H5_V1.1.1` to the destination folder. The source folder contains:

- _htmresc
- Projects
- Additional_License_Terms_SMAK.html
- Package_license_SMAK.html
- Release_Notes_SMAK.html

Below the file explorer, a dialog box titled `Replace or Skip Files` is shown. It displays the message: `Copying 19 items from X-Cube-SEC-M-H5_V... to STM32Cube_FW_H5_V...` and `The destination has 5 files with the same names`. The dialog offers three options:

- ☒ **Replace the files in the destination**
- ☐ **Skip these files**
- ☐ **Let me decide for each file**

Tools installation

- SW Requirements:
 - [STM32CubeMX](#) version 6.9.2 or upper 
 - [STM32CubeH5](#) version 1.1.1 
 - [STM32TRUSTEE-SM](#) version 1.1.1 
 - [STM32CubeIDE](#) version 1.13.2 or upper
 - [STM32CubeProgrammer](#) version 2.14.0 or upper
 - [Material for hands-on](#)
 - Terminal application (TeraTerm)

Tools installation : STM32CubeIDE

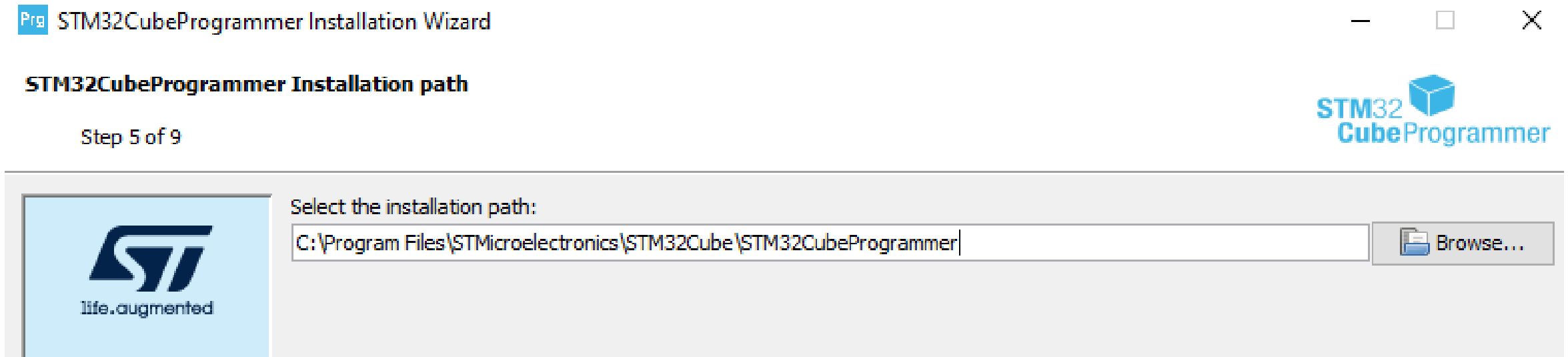
- [STM32CubeIDE](#) version 1.13.2 or upper
- Download and install the package.
- [STM32CubeIDE installation guide](#)

Tools installation

- SW Requirements:
 - [STM32CubeMX](#) version 6.9.2 or upper 
 - [STM32CubeH5](#) version 1.1.1 
 - [STM32TRUSTEE-SM](#) version 1.1.1 
 - [STM32CubeIDE](#) version 1.13.2 or upper 
 - [STM32CubeProgrammer](#) version 2.14.0 or upper
 - [Material for hands-on](#)
 - Terminal application (TeraTerm)

Tools installation : STM32CubeProgrammer

- [STM32CubeProgrammer](#) version 2.14.0 or upper
- Remark : please keep the default path installation if possible.



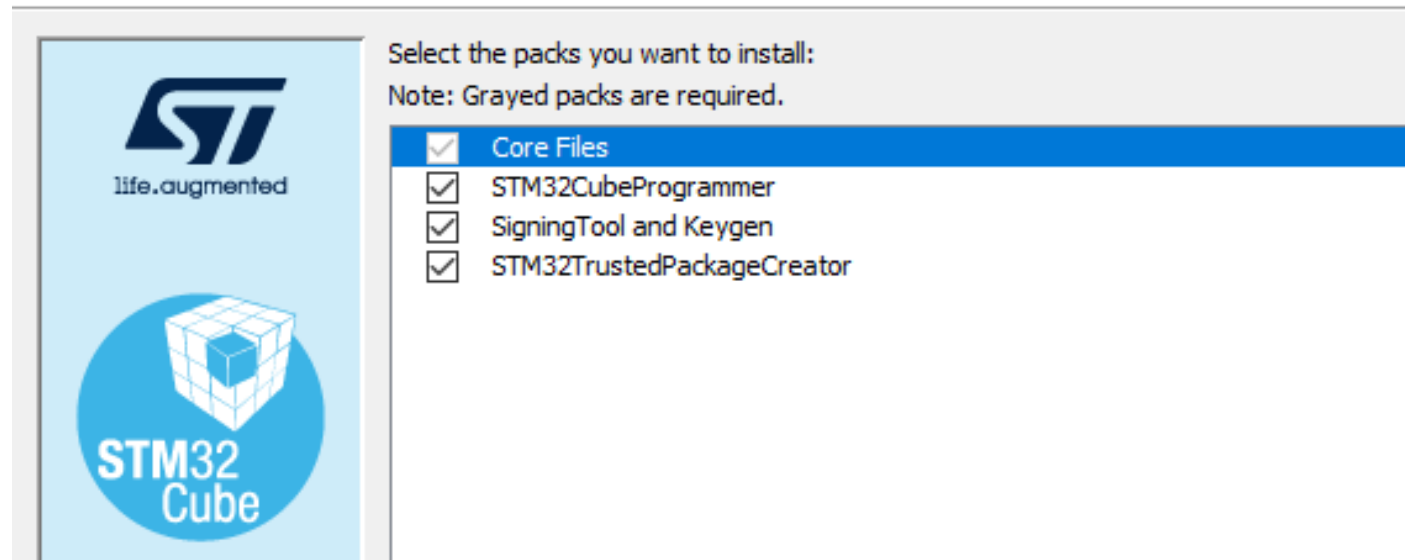
Tools installation : STM32CubeProgrammer

- Install all the components (default selection)

Prg STM32CubeProgrammer Installation Wizard

STM32CubeProgrammer Components selection

Step 6 of 9



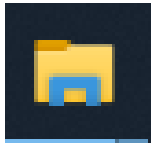
Tools installation : STM32CubeProgrammer

- If you haven't installed STM32CubeProgrammer in the default folder :

C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer

- Please edit the file :

C:\Users\<your_login>\STM32Cube\Repository\STM32Cube_FW_H5_V1.1.1\Projects\STM32H573I-DK\ROT_Provisioning\env.bat



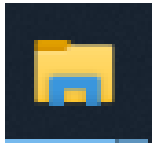
Tools installation : STM32CubeProgrammer

Update those variable with your installation path

\\STM32Cube\Repository\STM32Cube_FW_H5_V1.1.1\Projects\STM32H573I-DK\ROT_Provisioning\env.bat - Notepad++

View Encoding Language Settings Tools Macro Run Plugins Window ?

```
1 @ECHO OFF
2
3 :: =====
4 :: ..... General
5 :: =====
6 :: Configure tools installation path
7 set stm32programmercli="C:\Program
Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer\bin\STM32_Programmer_CLI.exe"
8 set stm32tpccli="C:\Program
Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer\bin\STM32TrustedPackageCreator_CLI.exe"
9
10 :: =====
11 :: ..... STiRoT boot path
12 :: =====
```



Tools installation : STM32CubeProgrammer

- If you haven't installed STM32CubeProgrammer in the default folder :

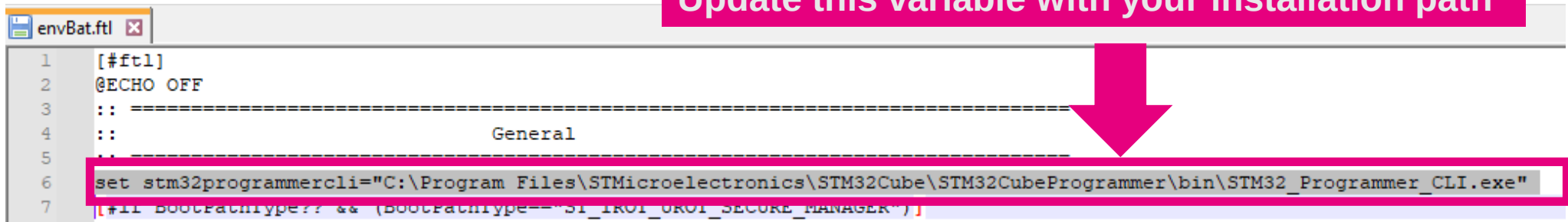
C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer

- Please edit the file :

C:\Program

Files\STMicroelectronics\STM32Cube\STM32CubeMX\db\plugins\projectmanager\envBat.ftl

Update this variable with your installation path



```
1  [#ftl]
2  @ECHO OFF
3  :: =====
4  ::                               General
5  :: =====
6  set stm32programmercli="C:\Program Files\STMicroelectronics\STM32Cube\STM32CubeProgrammer\bin\STM32_Programmer_CLI.exe"
7  [!if %BOOTPATHTYPE% == "ST_IK01_0K01_SECURE_MANAGER"]
```

Tools installation

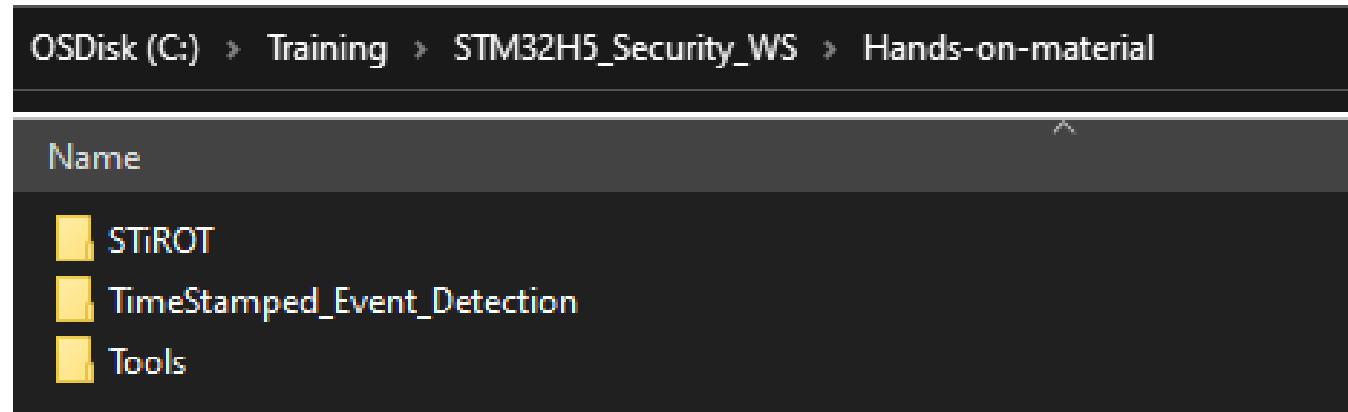
- SW Requirements:
 - [STM32CubeMX](#) version 6.9.2 or upper 
 - [STM32CubeH5](#) version 1.1.1 
 - [STM32TRUSTEE-SM](#) version 1.1.1 
 - [STM32CubeIDE](#) version 1.13.2 or upper 
 - [STM32CubeProgrammer](#) version 2.14.0 or upper 
 - [Material for hands-on](#)
 - Terminal application (TeraTerm)

Tool installation : Material for hands-on

- Download the archive : STM32H5_Security_WS_V1.0.zip available thanks this link: [Material for hands-on](https://drive.google.com/file/d/18Qv_AIPIp1u_uQBaQZE3_j8byiplT8Rz/view?usp=sharing)

https://drive.google.com/file/d/18Qv_AIPIp1u_uQBaQZE3_j8byiplT8Rz/view?usp=sharing

- Unzip this archive on your local drive.
Suggested path : “C:\Training\STM32H5_Security_WS”
- Expected result :



Tools installation

- SW Requirement :
 - [STM32CubeMX](#) version 6.9.2 or upper 
 - [STM32CubeH5](#) version 1.1.1 
 - [STM32TRUSTEE-SM](#) version 1.1.1 
 - [STM32CubeIDE](#) version 1.13.2 or upper 
 - [STM32CubeProgrammer](#) version 2.14.0 or upper 
 - [Material for hands-on](#) 
 - Terminal application (TeraTerm)

Tools installation : Terminal application

- Terminal application used in this workshop is TeraTerm
- You should be able to get TeraTerm from [here](#)
- Get the zip file and extract it to your preferred directory
- For convenience you can change some default settings
 - To get default baudrate to 115200, you can edit the TERATERM.INI file and change the line BaudRate=9600 to BaudRate=115200
 - To get popup window selecting com port by default, change Port=tcpip to Port=Serial in the same configuration file.

Tools installation

- SW Requirement :

- [STM32CubeMX](#) version 6.9.2 or upper 
- [STM32CubeH5](#) version 1.1.1 
- [STM32TRUSTEE-SM](#) version 1.1.1 
- [STM32CubeIDE](#) version 1.13.2 or upper 
- [STM32CubeProgrammer](#) version 2.14.0 or upper 
- [Material for hands-on](#) 
- Terminal application (TeraTerm) 

Sanity check

Sanity check.



Please check if you can find the file : SecureManagerPackage_PROD_v1.0.0.ssfi at this location :

C:\Users\<your_login>\STM32Cube\Repository\STM32Cube_FW_H5_V1.1.1\Projects\STM32H573I-DK\ROT_Provisioning\SM\Binary\



Please compile the CubeIDE project SMAK_Appli :

C:\Users\<your_login>\STM32Cube\Repository\STM32Cube_FW_H5_V1.1.1\Projects\STM32H573I-DK\Applications\ROT\SMAK_Appli\STM32CubeIDE

If you are not familiar with STM32CubeIDE follows the instruction in the next slides.

Sanity check.



Please check if you can find the file : SecureManagerPackage_PROD_v1.1.1.ssfi at this location :

C:\Users\<your_login>\STM32Cube\Repository\STM32Cube_FW_H5_V1.1.1\Projects\STM32H573I-DK\ROT_Provisioning\SM\Binary\

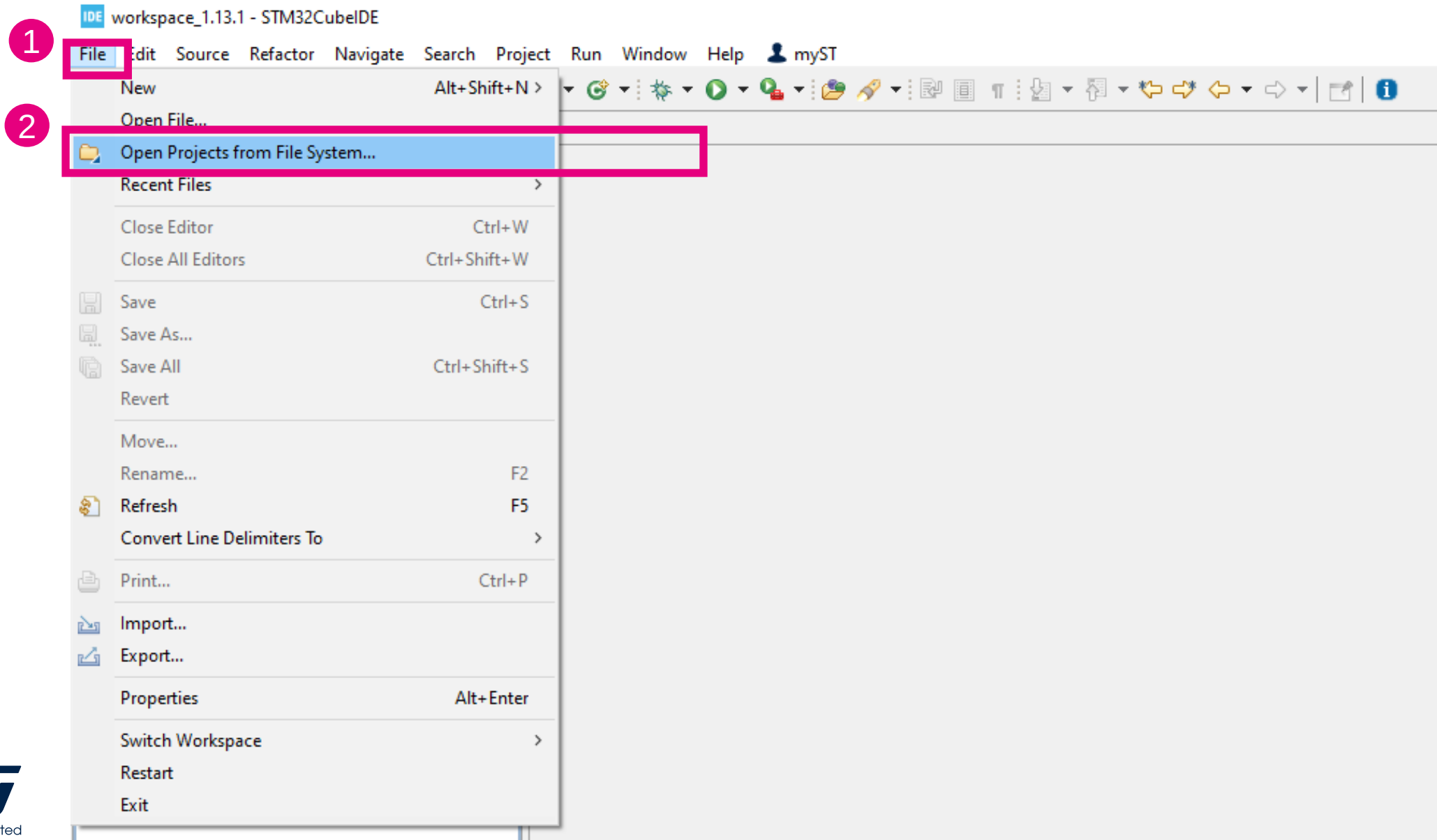


Please compile the CubeIDE project SMAK_Appli :

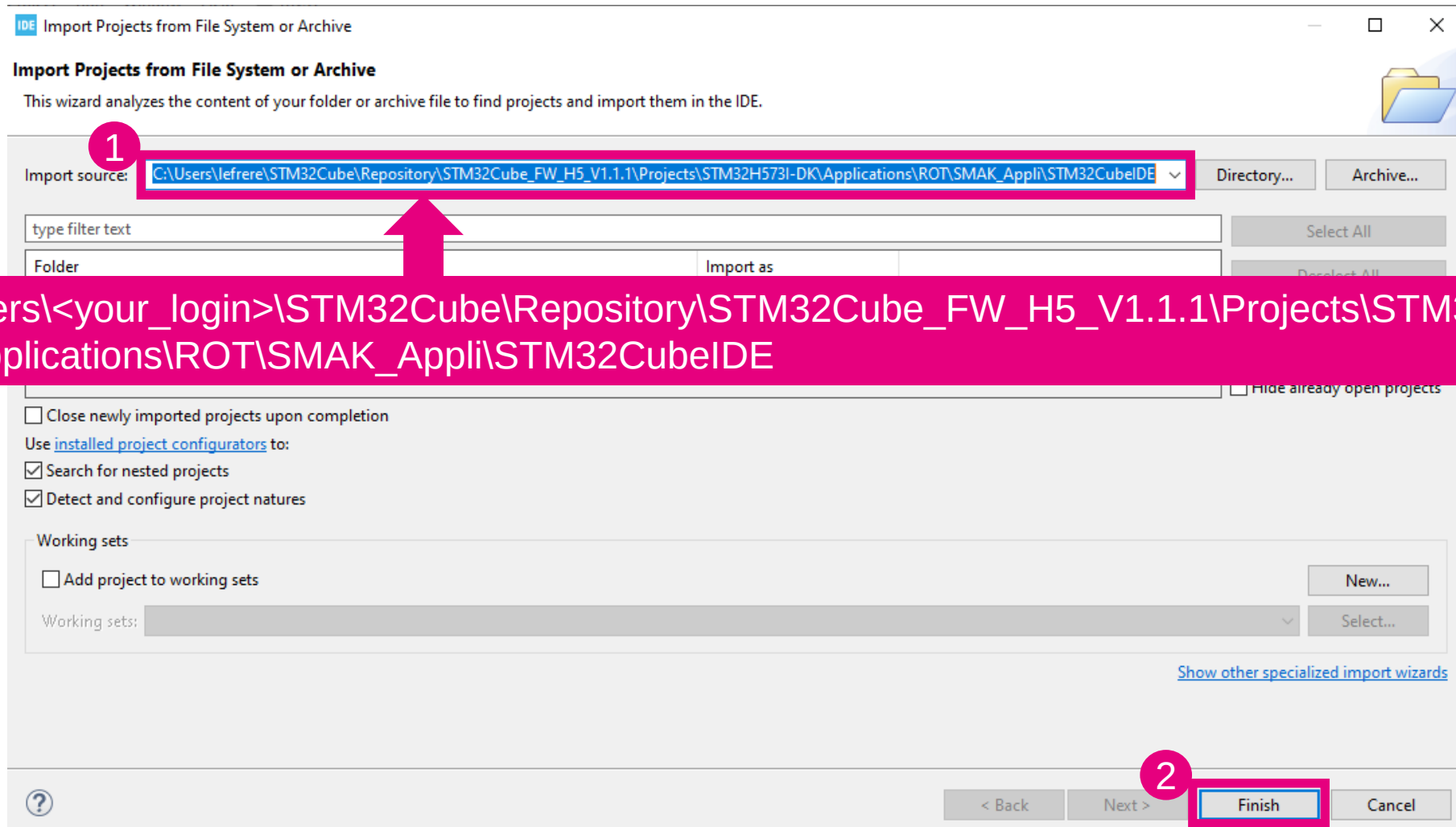
C:\Users\<your_login>\STM32Cube\Repository\STM32Cube_FW_H5_V1.1.1\Projects\STM32H573I-DK\Applications\ROT\SMAK_Appli\STM32CubeIDE

If you are not familiar with STM32CubeIDE follows the instruction in the next slides.

Import SMAK application in the IDE



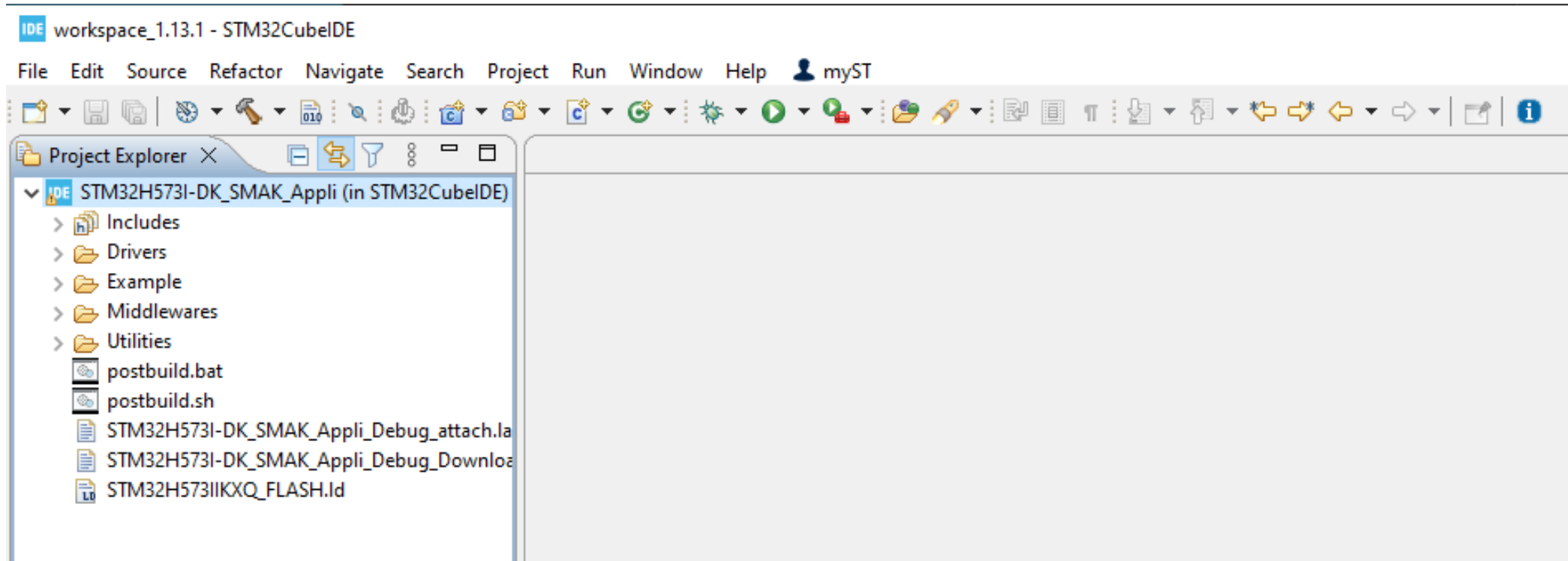
Import SMAK application in the STM32CubeIDE



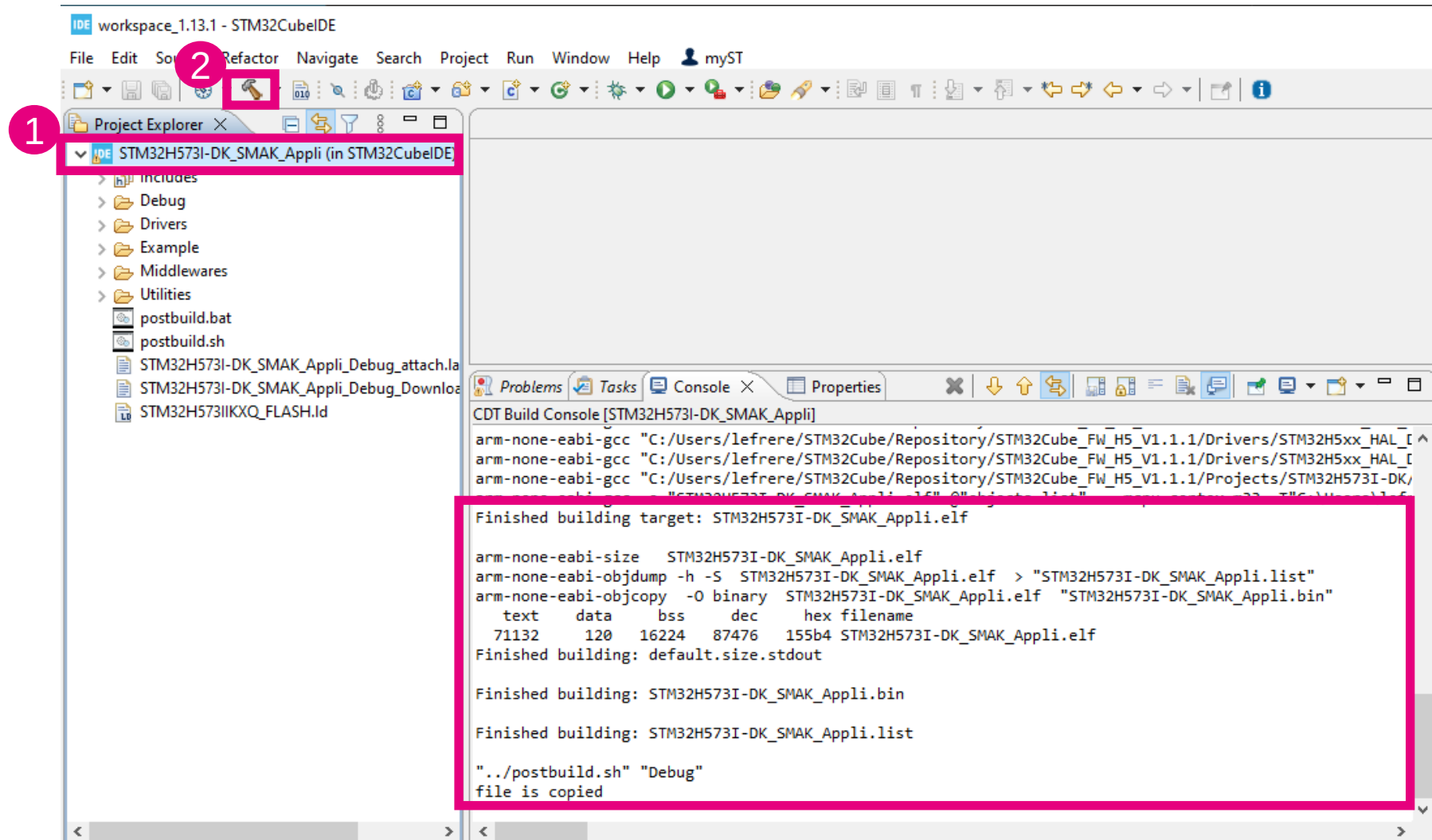
C:\Users\<your_login>\STM32Cube\Repository\STM32Cube_FW_H5_V1.1\Projects\STM32H573I-DK\Applications\ROT\SMAK_Appli\STM32CubeIDE



Import SMAK application in the STM32CubeIDE



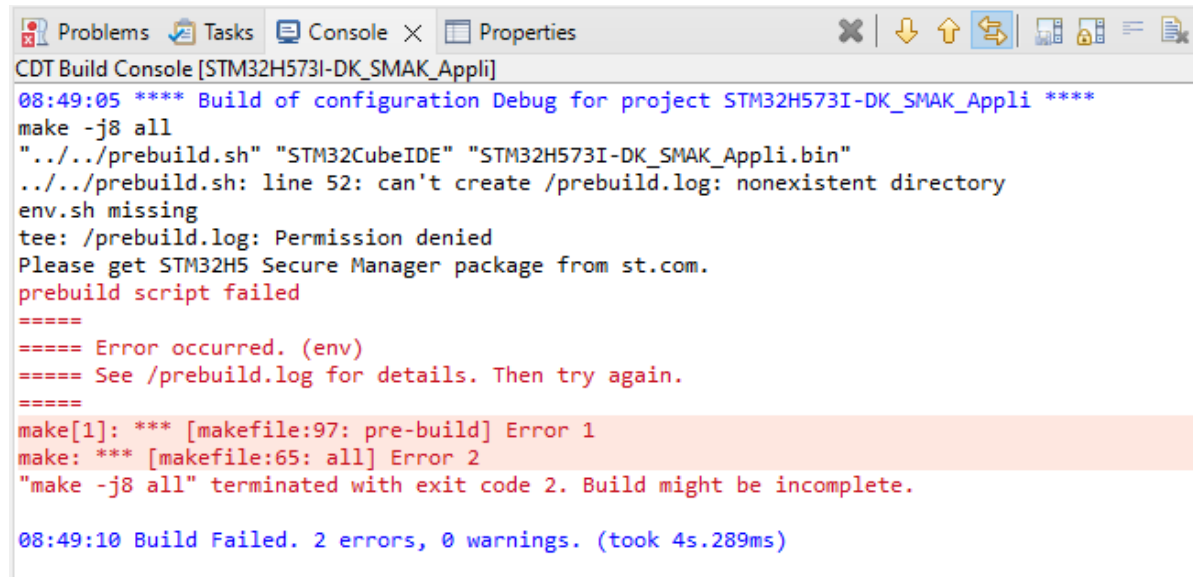
Compile SMAK application with STM32CubeIDE



**Issue with STM32CubeIDE 1.14.0
Fixed in 1.14.1 !**

Issue with STM32CubeIDE 1.14.0

- If you are using STM32CubeIDE 1.14.0, make application failed.



```
Problems Tasks Console X Properties
CDT Build Console [STM32H573I-DK_SMAK_Appli]
08:49:05 **** Build of configuration Debug for project STM32H573I-DK_SMAK_Appli ****
make -j8 all
"../../prebuild.sh" "STM32CubeIDE" "STM32H573I-DK_SMAK_Appli.bin"
../../prebuild.sh: line 52: can't create /prebuild.log: nonexistent directory
env.sh missing
tee: /prebuild.log: Permission denied
Please get STM32H5 Secure Manager package from st.com.
prebuild script failed
=====
===== Error occurred. (env)
===== See /prebuild.log for details. Then try again.
=====
make[1]: *** [makefile:97: pre-build] Error 1
make: *** [makefile:65: all] Error 2
"make -j8 all" terminated with exit code 2. Build might be incomplete.

08:49:10 Build Failed. 2 errors, 0 warnings. (took 4s.289ms)
```

Issue with STM32CubeIDE 1.14.0

- Extract postbuild.7z available in this post :

<https://community.st.com/t5/stm32-trust-tee-secure-manager/i-cannot-compile-smak-application/m-p/614812/highlight/true#M45>

- in this folder (it will replace the current postbuild.bat)

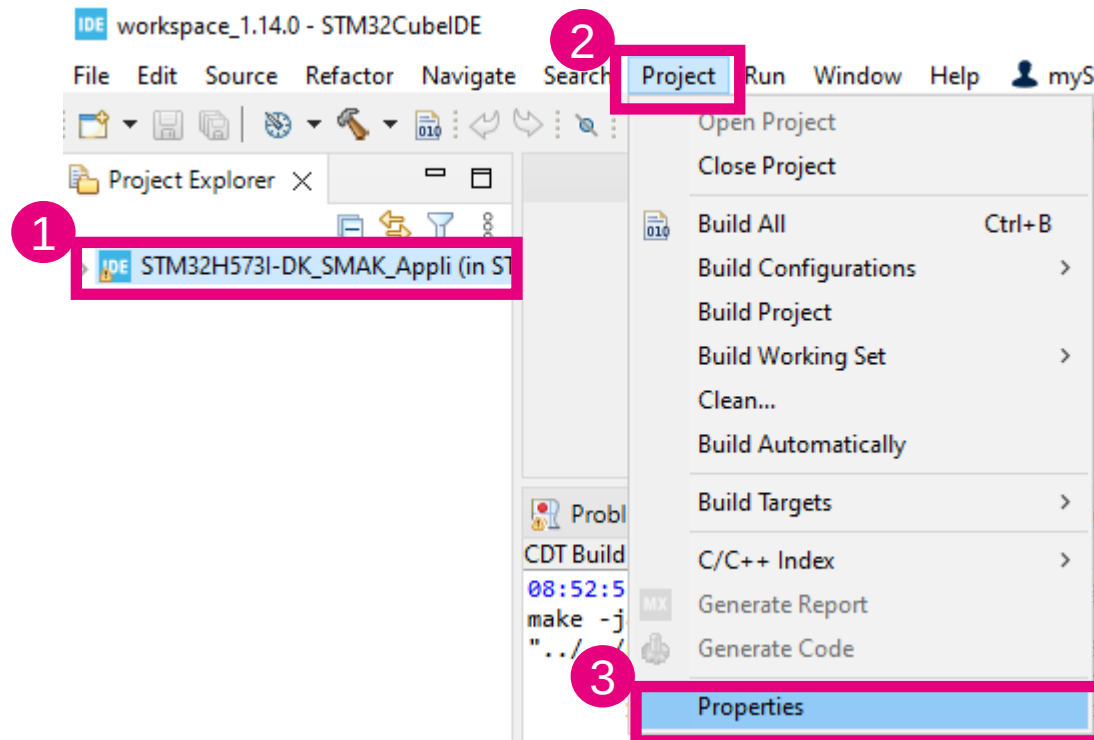
C:\Users\<your_login>\STM32Cube\Repository\STM32Cube_FW_H5_V1.1.1\Projects\STM32H573I-DK\Applications\ROT\SMAK_Appli\STM32CubeIDE\

This issue only concerns version 1.14.0 !

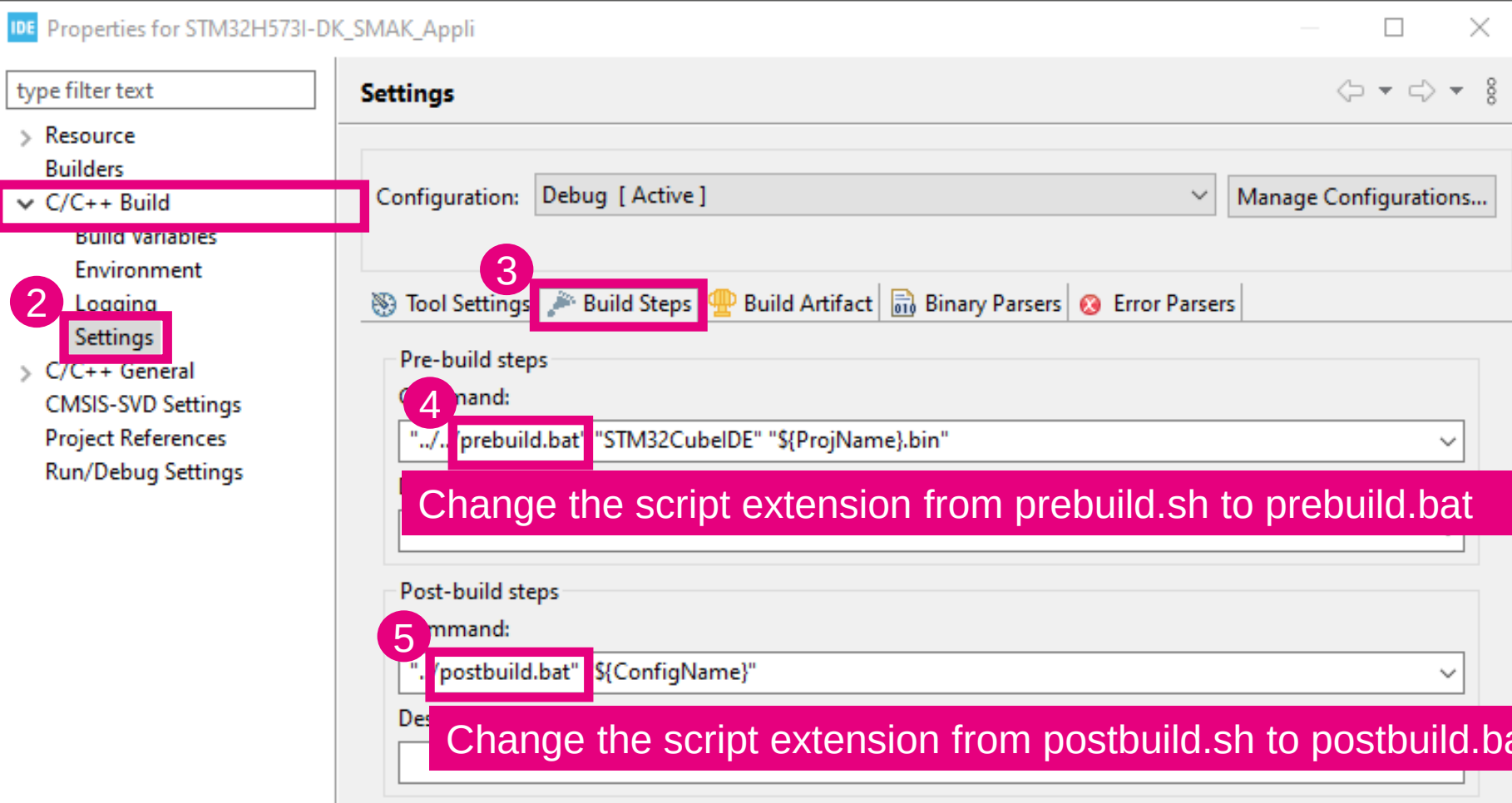
Nothing to be done on 1.14.1

Issue with STM32CubeIDE 1.14.0

- Then in STM32H573I-DK_SMAK_Appli project properties



Issue with STM32CubeIDE 1.14.0



The screenshot shows the 'Settings' dialog for the 'C/C++ Build' configuration. The left sidebar shows the 'Settings' option selected under 'C/C++ Build'. The main panel shows the 'Build Steps' tab. The 'Pre-build steps' section has a 'Command' field containing the path to 'prebuild.bat'. The 'Post-build steps' section has a 'Command' field containing the path to 'postbuild.bat'. A pink callout box highlights the 'prebuild.bat' path with the text 'Change the script extension from prebuild.sh to prebuild.bat'. Another pink callout box highlights the 'postbuild.bat' path with the text 'Change the script extension from postbuild.sh to postbuild.bat'. The 'Apply and Close' button is highlighted with a pink callout box and the number 6.

1. C/C++ Build

2. Settings

3. Build Steps

4. Command: `"../prebuild.bat" "STM32CubeIDE" "${ProjName}.bin"`

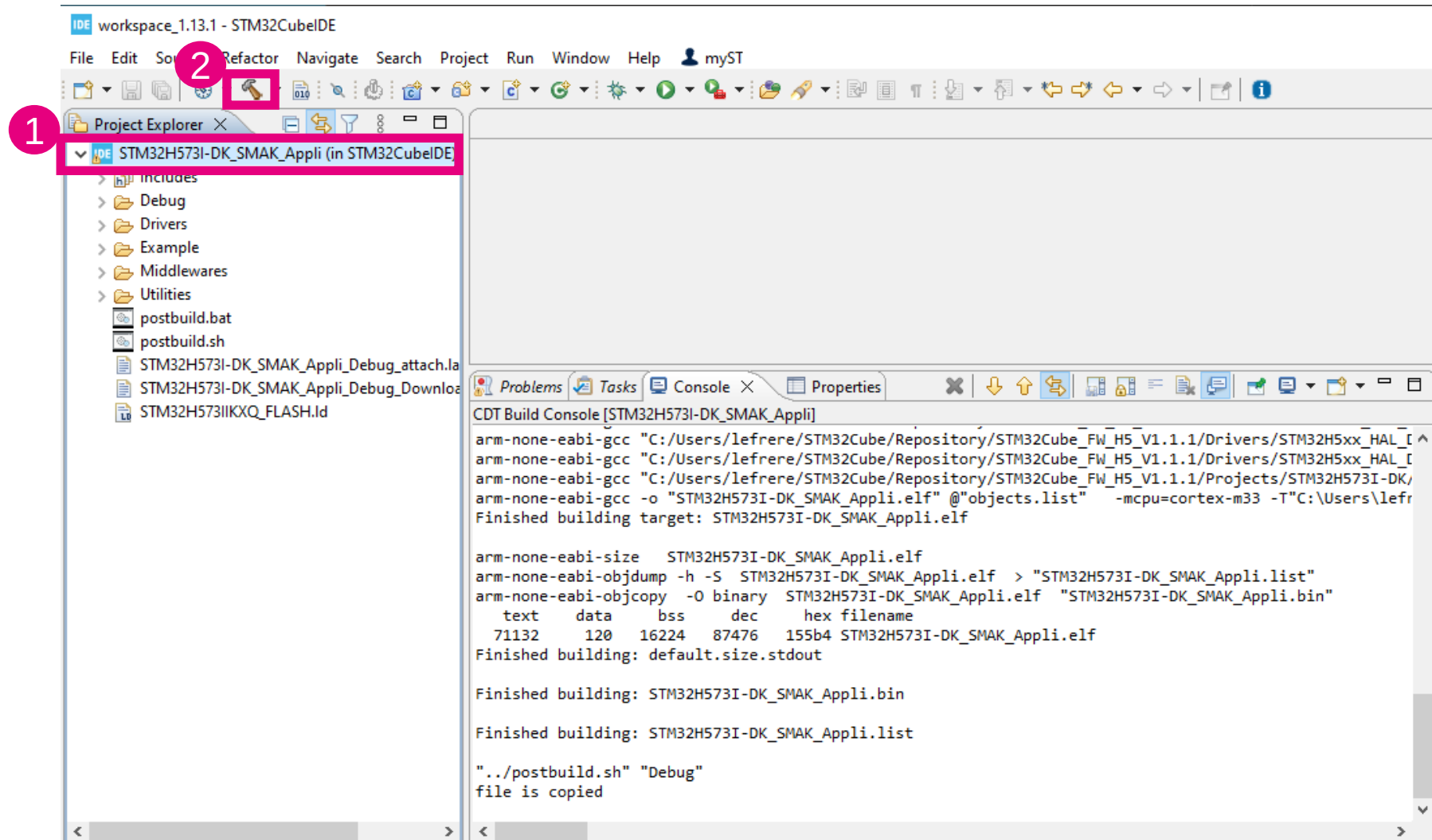
Change the script extension from prebuild.sh to prebuild.bat

5. Command: `"../postbuild.bat" ${ConfigName}"`

Change the script extension from postbuild.sh to postbuild.bat

6. Apply and Close

Compile SMAK application with STM32CubeIDE



Sanity check done.

Congratulation !

You're ready...

We would be please to meet you at
the STM32H5 Security workshop !

Our technology starts with You



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